

Linear regression II: basis functions and collinearity

Fit curves with a model that stays linear in θ , see why dependent columns spoil $\hat{\theta}$, and encode categories without the dummy variable trap.

THE QUESTION

Least squares fits a straight line. How do we fit curves with it, and what goes wrong when the columns of X depend on each other?

THE INTUITION

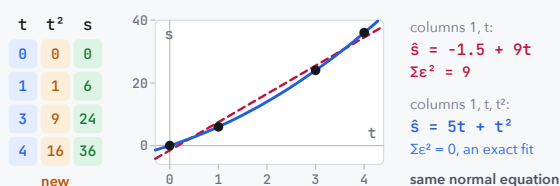
Put features $\varphi(x)$ such as powers, sines or bumps in the columns instead of x , and solve the same normal equation: the model is still linear in θ . If one column is (nearly) a combination of the others, many θ give (nearly) the same fit, so the data cannot pin $\hat{\theta}$ down.

KEY TAKEAWAY

"Linear" means linear in the parameters. Keep the columns independent: drop copies, scale inputs, and code N categories with $N - 1$ dummies.

input $x \rightarrow$ features $\varphi(x) = (1, x, x^2, \dots) \rightarrow$ design matrix $\Phi \rightarrow \theta = (\Phi^T \Phi)^{-1} \Phi^T y \rightarrow$ check rank and $\text{cond}(\Phi^T \Phi)$

1 Add t^2 as a new column



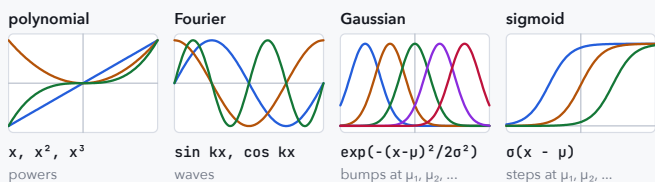
The lecture's table gives distance s at times t . The best line leaves $\sum \epsilon_i^2 = 9$. Add the column t^2 and fit $\hat{s} = \theta_0 + \theta_1 t + \theta_2 t^2$ with the same normal equation, now with rows $(1, t_i, t_i^2)$: the fit $\hat{s} = 5t + t^2$ is exact. Other useful transformations are $\log x$ and products such as $x_1 x_2$.

2 Linear means linear in the parameters

$\hat{s} = \theta_0 + \theta_1 t$	features 1, t	linear in θ
$\hat{s} = \theta_0 + \theta_1 t + \theta_2 t^2$	features 1, t, t ²	linear in θ
$\hat{s} = \theta_0 + \theta_1 t + \theta_2 t^2 + \theta_3 \cos(t^3)$	features 1, t, t ² , $\cos(t^3)$	linear in θ
$\hat{s} = \theta_0 + \theta_1 t + \exp(\theta_2) \cdot t$	θ_2 sits inside exp	not linear

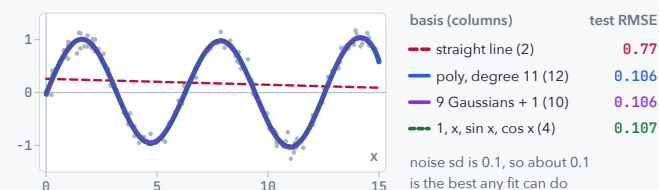
In the first three models each θ_j multiplies a fixed function of the input, so $\hat{s} = \Phi \theta$ with one column per function. The fourth is not linear regression: e^{θ_2} is not a constant times θ_2 . A quick test: $\partial \hat{y} / \partial \theta_j$ must not depend on θ . The shape in t can be anything.

3 A basis function turns x into K features



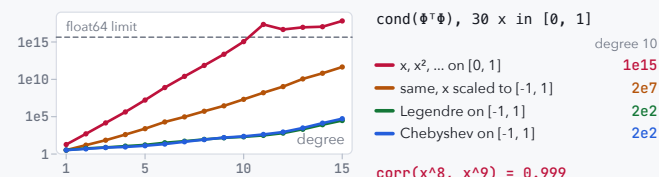
$\varphi: \mathbb{R}^D \rightarrow \mathbb{R}^K$ maps each input to K numbers and the model is $\hat{y} = \theta^T \varphi(x)$. Stack the rows $\varphi(x_i)^T$ into Φ ; then $\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$. Common families, each with the constant 1: powers x^k ; Fourier $\sin kx, \cos kx$; Gaussian bumps $\exp(-(x - \mu_j)^2 / 2\sigma^2)$; sigmoids $\sigma(x - \mu_j)$ with $\sigma(z) = 1 / (1 + e^{-z})$. The centres μ_j and the width σ are chosen, not fitted. Random Fourier features draw the frequencies at random.

4 Three bases fit the same wiggly data



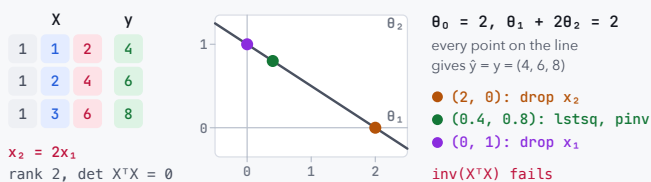
The basis notebook's data: $y = \sin x$ plus noise; dots are the 160 training points. A line fails. Polynomials (x scaled to $[-1, 1]$), Gaussian bumps and sines all reach the noise floor, with 12, 10 and 4 columns: a basis that matches the shape needs fewer. Counts were picked on a held-out 20%. The test error is flat near the floor, so the pick is fragile: use cross-validation.

5 Powers of x are nearly collinear columns



On $[0, 1]$, x^8 and x^9 are almost the same column, and $\kappa = \lambda_{\max} / \lambda_{\min}$ of $\Phi^T \Phi$ hits the float64 limit by degree 10. On the polynomial notebook's 30 points, degree 13 has $|\theta_j|$ up to $4 \cdot 10^8$ and degree 30 a test RMSE above 5000. Scaling x to $[-1, 1]$ helps; Legendre or Chebyshev polynomials span the same functions with κ near 200.

6 A copied column makes $X^T X$ singular

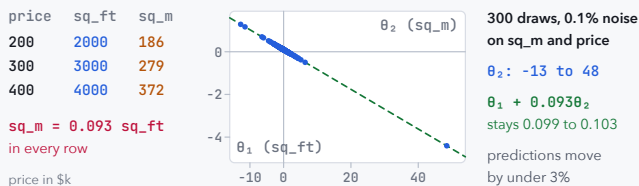


The notebook's example has $x_2 = 2x_1$: rank 2, not 3, and $\det(X^T X) = 0$, the singular case of the linear regression I sheet. As $\theta_1 x_1 + \theta_2 x_2 = (\theta_1 + 2\theta_2)x_1$, every θ on the line fits. np.linalg.lstsq returns the shortest, (2, 0.4, 0.8); dropping x_2 gives (2, 2).

Near-dependent columns make $\hat{\theta}$ unstable

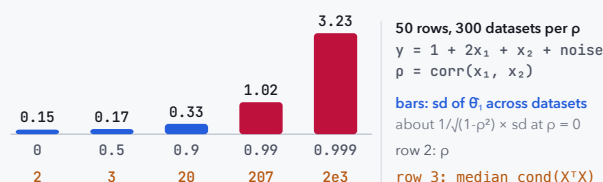
Correlated features leave the predictions intact but scatter the coefficients. A category with N levels needs $N - 1$ indicator columns and a reference level.

7 Near-copies scatter the coefficients



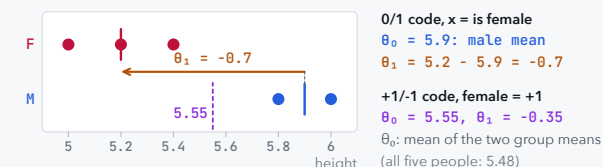
Multicollinearity: a feature is a linear combination of others, exactly ($\det(X^T X) = 0$, panel 6) or nearly ($\det \approx 0$). The lecture's houses record area in sq_ft and in sq_m. The columns are proportional, so any θ with $\theta_1 + 0.093\theta_2 = 0.1$ fits the clean data. With 0.1% noise the fit is unique but fragile: over 300 noisy copies θ_2 runs from -13 to 48, while $\theta_1 + 0.093\theta_2$ and the predictions barely move. The coefficients cannot say which feature matters.

8 The spread of $\hat{\theta}_j$ grows like $1/\sqrt{1-\rho^2}$



Two features with correlation ρ and the same true θ . At $\rho = 0.999$ the standard deviation of $\hat{\theta}_1$ is 22 times its value at $\rho = 0$ and $\kappa(X^T X)$ is about 2000; the average stays near the true 2. The lecture's fixes: drop one of the pair, or regularise (ridge adds λI to $X^T X$; see the ridge regression sheet). Check `np.linalg.cond(X.T @ X)` before reading a coefficient.

11 Dummy coefficients compare group means



The lecture's heights with $x = 1$ for female: $\theta_0 = 5.9$ is the mean of the reference group and $\theta_1 = -0.7$ is the female mean minus the male mean. With the ± 1 code (two categories only), $\theta_1 = -0.35$ is half the gap and $\theta_0 = 5.55$ the average of the two group means, not the mean of all five heights (5.48).

9 Numbering categories invents an order



Coding wind direction N, E, W, S as 0, 1, 2, 3 claims $S > W > E > N$ with equal steps, so the model can only put a straight line through the codes. In our illustrative table the true offsets are in no such order: the ordinal fit has RMSE 8.0; one indicator per direction gets 3.9.

10 Use $N - 1$ dummies and a reference

	1	N?	E?	W?		1	N?	E?	W?	S?		rank on the 60 rows (with vehicles, wind)
N	1	1	0	0	N	1	1	0	0	0		1 + 3 dummies 6 of 6
E	1	0	1	0	E	1	0	1	0	0		1 + 4 dummies 6 of 7
W	1	0	0	1	W	1	0	0	1	0		4, no 1s column 6 of 6
S	1	0	0	0	S	1	0	0	0	1		

S is the reference (all three are 0)

$N? + E? + W? + S? = 1$ the sum copies the 1s column

binary codes 00 to 11: fake shared bits

Give every category but one a 0/1 column. The left-out one (South) is the reference: each dummy coefficient is a difference from it. All N dummies add up to the ones column, so with an intercept X loses a rank and $X^T X$ is singular: the dummy variable trap. Binary codes use $\log_2 N$ columns but make categories share bits for no reason. Make the most common category the reference.

```
D = np.eye(4)[code] # N, E, W, S
X = np.column_stack([np.ones(60), D])
np.linalg.matrix_rank(X) # 4 of 5: trap
np.linalg.matrix_rank(X[:, :-1]) # 4 of 4
```

12 Pause and think

Is $\hat{y} = \theta_0 + \theta_1 \log x + \theta_2 x_1 x_2$ linear regression? Yes: it is linear in θ , with features $\log x$ and $x_1 x_2$.

Five categories and an intercept: how many dummies? Four. The fifth is the reference.

What if $x_3 = 3x_1 + 1$ exactly? x_3 is a combination of x_1 and the ones column, so $X^T X$ is singular.

Correlated features: are the predictions bad? No. The predictions are stable; single coefficients are not.

Degree 12 on raw x in $[0, 1]$: why the trouble? The powers are nearly collinear and κ passes 10^{16} . Scale x or use Legendre polynomials.

linear in θ , not in x · dependent columns $\rightarrow$ (nearly) singular $\Phi^T \Phi \rightarrow$ unstable $\hat{\theta}$ · N categories $\rightarrow N - 1$ dummies

Before: the linear regression I sheet; next: the gradient descent sheet. Source: lecture slides "Linear regression" and the basis, basis2, polynomial-features and dummy-variables-multi-collinearity notebooks (nipunbatra.github.io/ml-teaching). The data split, the correlation simulation, the noise draws and the pollution table are ours, computed with fixed seeds.